

Task	Weighting condition	Qwen4	Qwen9	Gemma	Llama
SH-Doc	Unit weights	94.00	93.00	88.00	87.00
	Unit + key-fact boost	95.00	94.00	90.00	88.00
	AMOR	93.00	93.00	89.00	86.00
	AMOR + key-fact boost	95.00	94.00	90.00	88.00
MH-Doc	Unit weights	60.00	63.00	54.00	58.00
	Unit + key-fact boost	62.00	62.00	54.00	60.00
	AMOR	59.00	63.00	54.00	61.00
	AMOR + key-fact boost	61.00	62.00	53.00	61.00
FC-SH	Unit weights	75.00	82.00	71.00	45.00
	Unit + key-fact boost	78.00	82.00	74.00	42.00
	AMOR	75.00	82.00	71.00	43.00
	AMOR + key-fact boost	78.00	82.00	74.00	44.00
FC-MH	Unit weights	14.00	16.00	10.00	6.00
	Unit + key-fact boost	14.00	16.00	8.00	5.00
	AMOR	14.00	16.00	10.00	6.00
	AMOR + key-fact boost	14.00	16.00	8.00	5.00
LoCoMo	Unit weights	49.56	55.84	38.44	49.76
	Unit + key-fact boost	49.97	56.28	38.34	49.71
	AMOR	49.63	55.83	38.42	49.60
	AMOR + key-fact boost	49.96	56.15	37.99	49.40
2Wiki	Unit weights	57.72	57.98	48.33	49.86
	Unit + key-fact boost	58.02	57.90	48.94	50.50
	AMOR	59.14	59.23	49.15	50.62
	AMOR + key-fact boost	59.17	59.23	48.99	51.17

Table 1: Full configured data and native answer metrics for the matched weighting comparison. Key-fact boost is the transferred released CatRAG operator, not the full system. LoCoMo is the aggregate of its native category scores; 2Wiki is answer F1; document and FC tasks use substring EM. Source connections, query initialization, and context construction are unchanged. All values are percentages.

Supporting Results: Matched Recommendation Controls

The tables below use the original evaluation configurations and all 3,386 questions. The manuscript’s main evidence and examples are provided separately in the two-column analysis.